混凝土抗压强度预测

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Github：https://github.com/maimution/deep_learning.git

1、 Dataset 该数据为混凝土抗压强度数据，共 1030 实例。8 输入变量，1 输出变量。

Columns: Cement (component 1) ：kg in a m3 mixture -- Input Variable

Blast Furnace Slag (component 2)：kg in a m3 mixture -- Input Variable

Fly Ash (component 3)：kg in a m3 mixture -- Input Variable

Water (component 4)：kg in a m3 mixture -- Input Variable

Superplasticizer (component 5)：kg in a m3 mixture -- Input Variable

Coarse Aggregate (component 6)：kg in a m3 mixture -- Input Variable

Fine Aggregate (component 7)：kg in a m3 mixture -- Input Variable

Age：Day (1~365) -- Input Variable

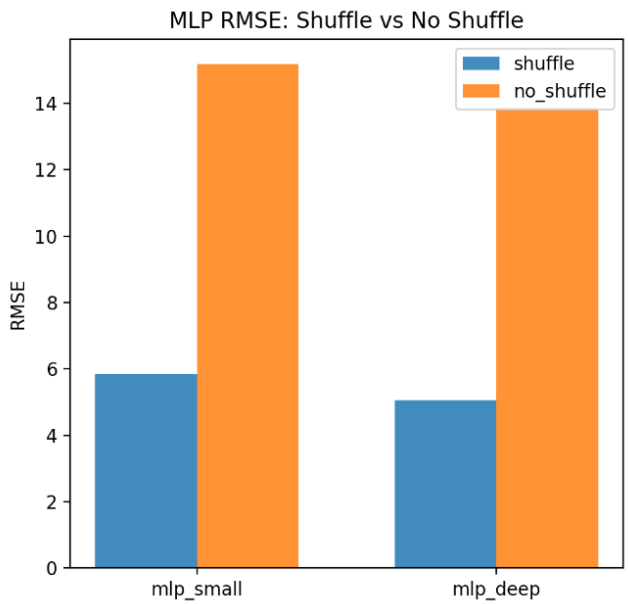
Concrete compressive strength：MPa -- Output Variable

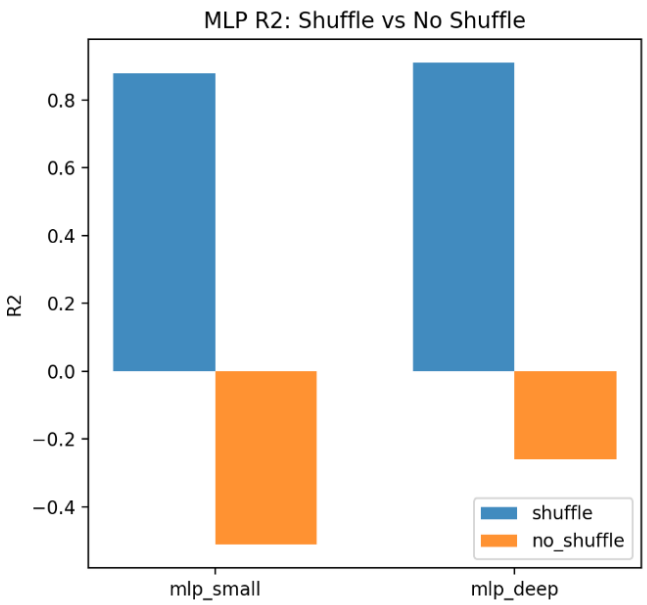
2、 采用的方法

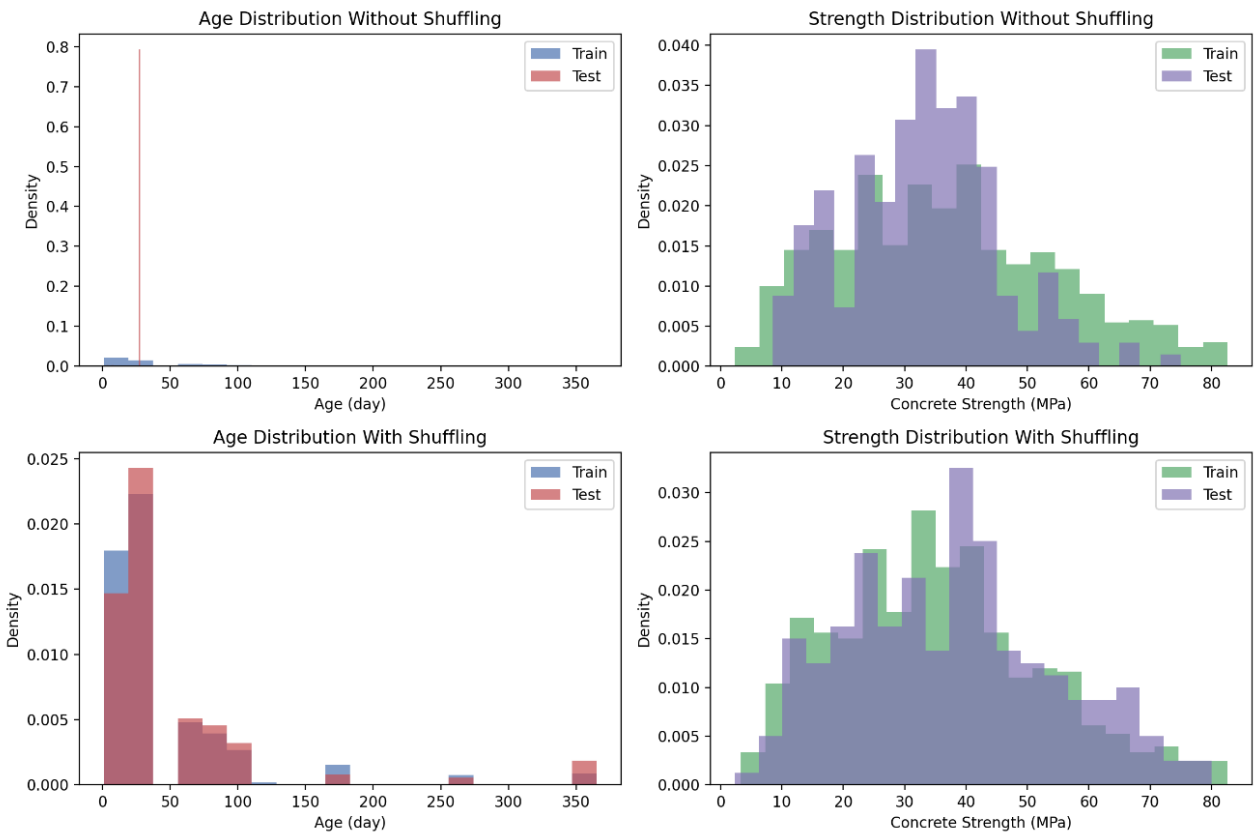
MLP_deep、MLP_small、linear_regression、pca_linear_regression、correlation_linear_regression

3、 数据集分析

如果不进行打乱，数据集的前 80%作为训练集后 20%作为测试集时，可以发现 Age 与 Mpa 的测试集训练集分布不太统一，所以进行打乱操作，否则会对 Mlp 的结果产生较大的影响，运行结果也确实如此。

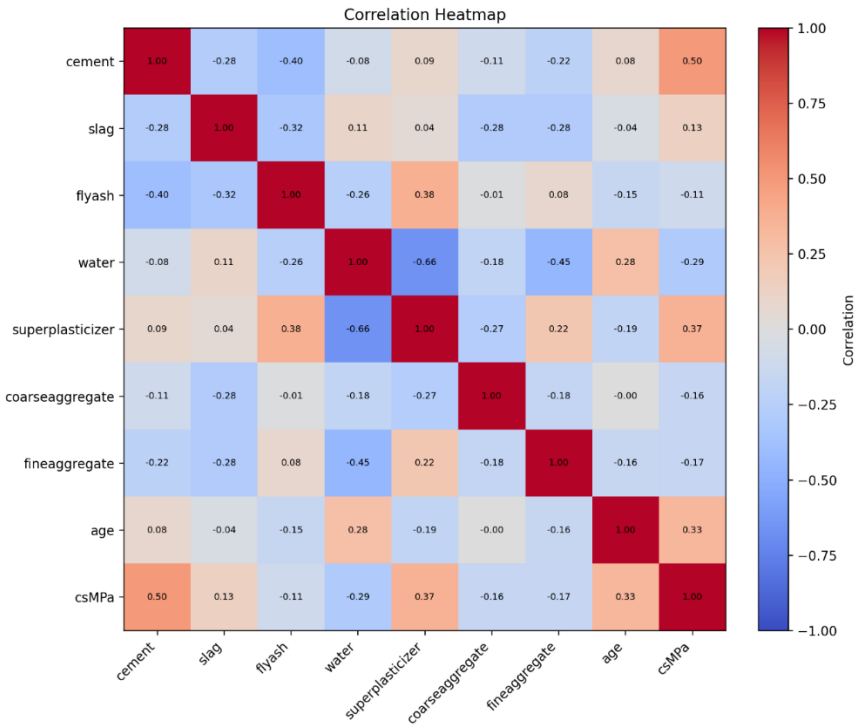


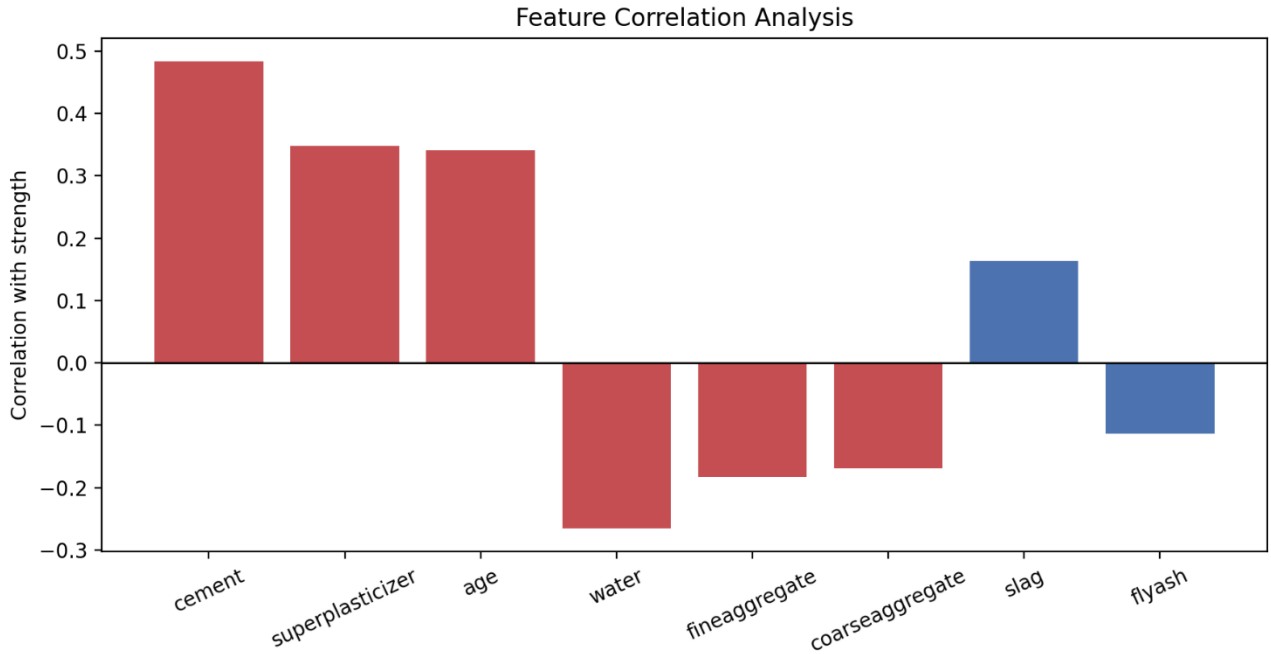




4、 相关性分析：

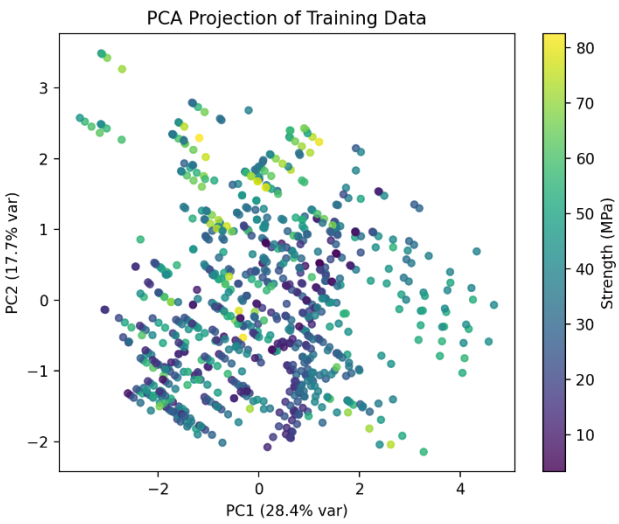
计算 Pearson 相关系数，算出每一列和别的列之间的相关性，取前 top_k=6 个特征作为输入；

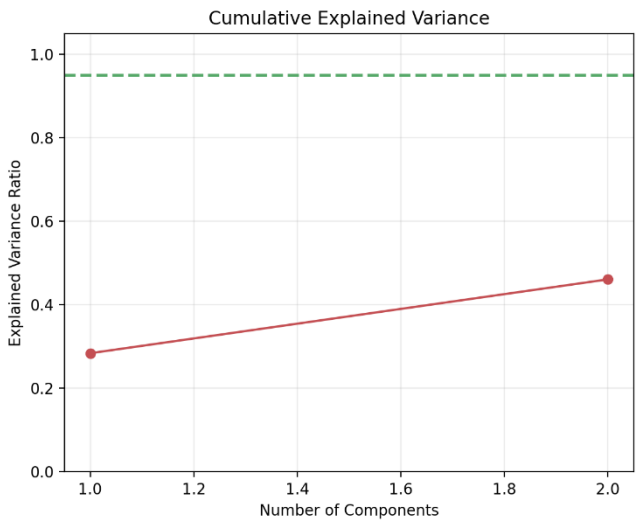




5、 主成分分析

先标准化，再用 PCA(n_components=0.95) 自动保留 95% 信息，再把降维后的主成分送入线性回归。为了方便可视化，这里只分析做 PCA(n_components=2)， 把训练集投影到二维平面，画散点图观察样本分布，实际代码用的是 PCA(n_components=0.95)。



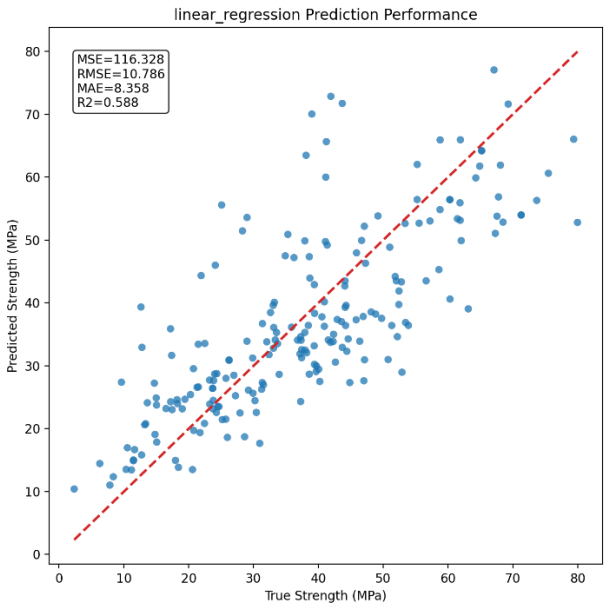


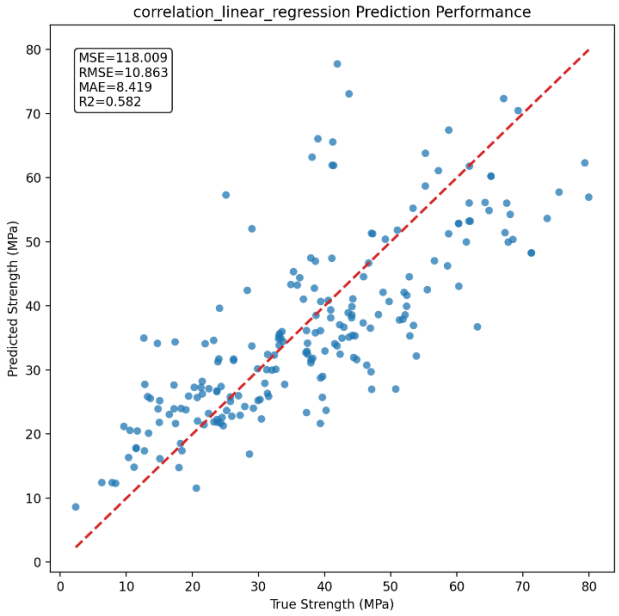

```
"pca_linear_regression": Pipeline(
    [
        ("scaler", StandardScaler()),
        ("pca", PCA(n_components=0.95)),
        ("model", LinearRegression()),
    ]
),
```

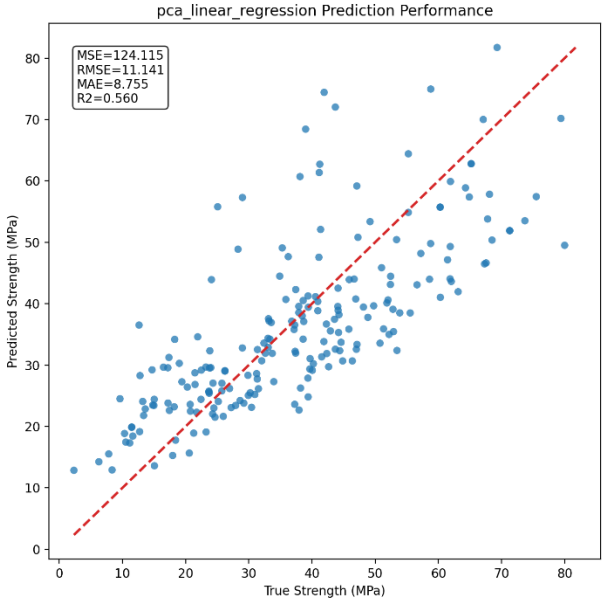
```
def save_pca_plot(x_train, y_train):
    scaler = StandardScaler()
    x_train_scaled = scaler.fit_transform(x_train)
    pca = PCA(n_components=2)
    components = pca.fit_transform(x_train_scaled)

    fig, axes = plt.subplots(1, 2, figsize=(12, 5))
```

6、 线性回归（全特征输入、PCA 降维度输入、top_6 特征输入）







7、 神经网络回归

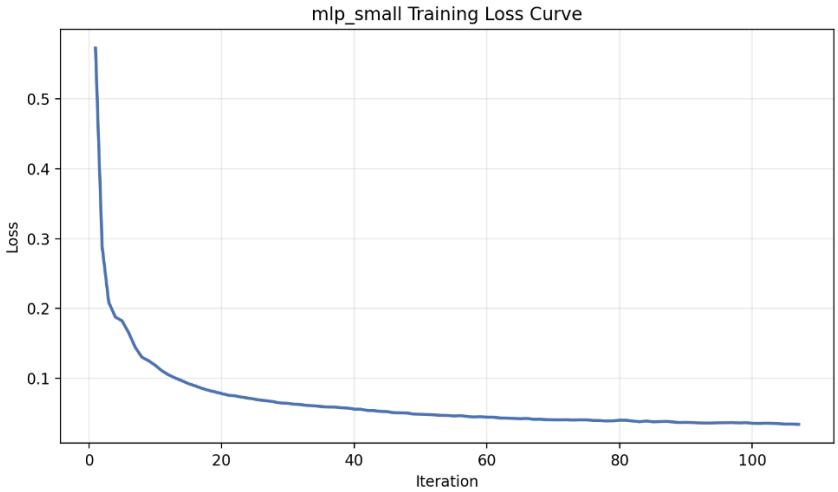
mlp_small

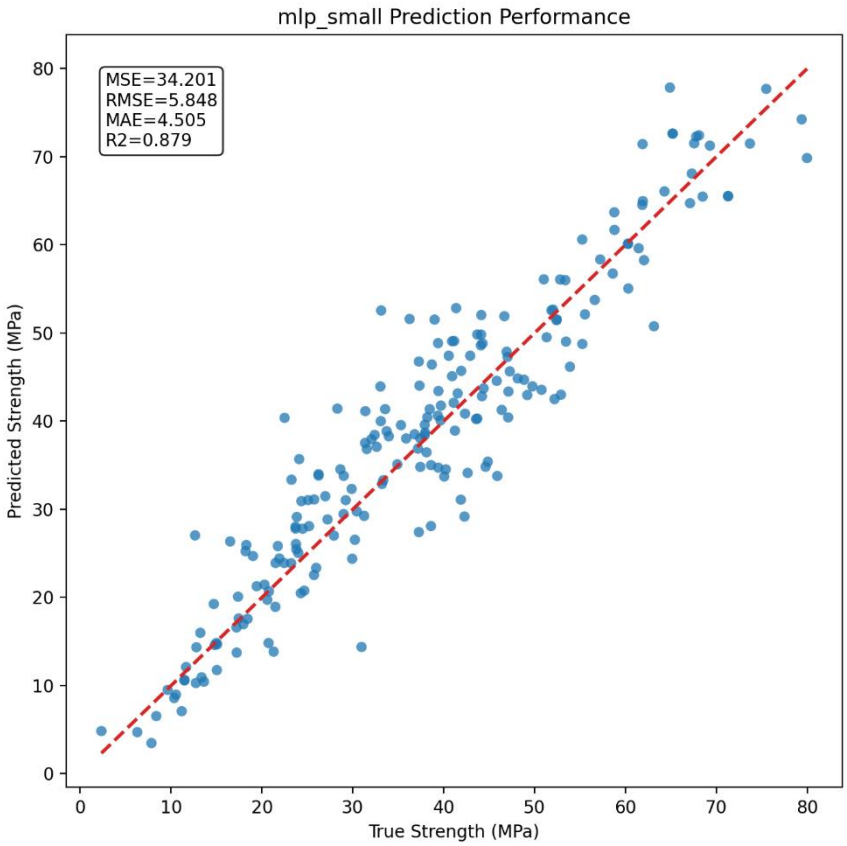
- hidden_layer_sizes=(32,)
- activation="relu"
- solver="adam"
- learning_rate_init=0.01
- max_iter=3000
- early_stopping=True
- validation_fraction=0.1
- random_state=RANDOM_STATE

mlp_deep

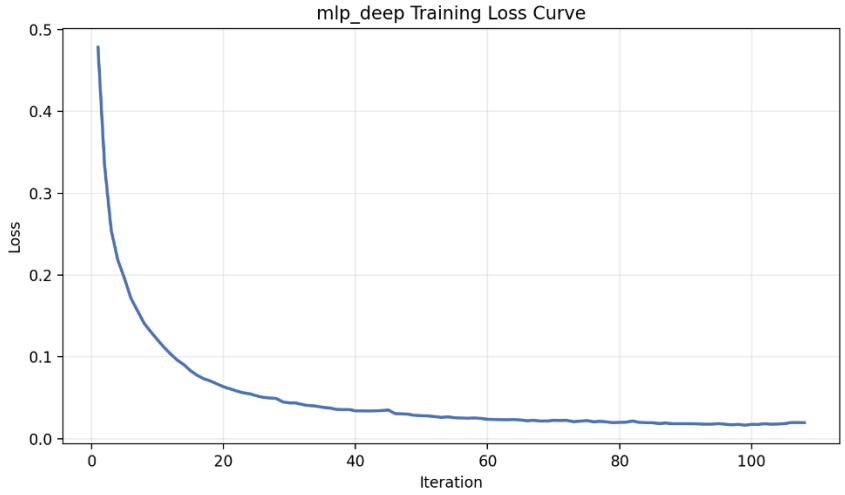
- hidden_layer_sizes=(64, 32, 16)
- activation="relu"
- solver="adam"
- alpha=1e-4
- learning_rate_init=0.003
- max_iter=4000
- early_stopping=True
- validation_fraction=0.1
- random_state=RANDOM_STATE

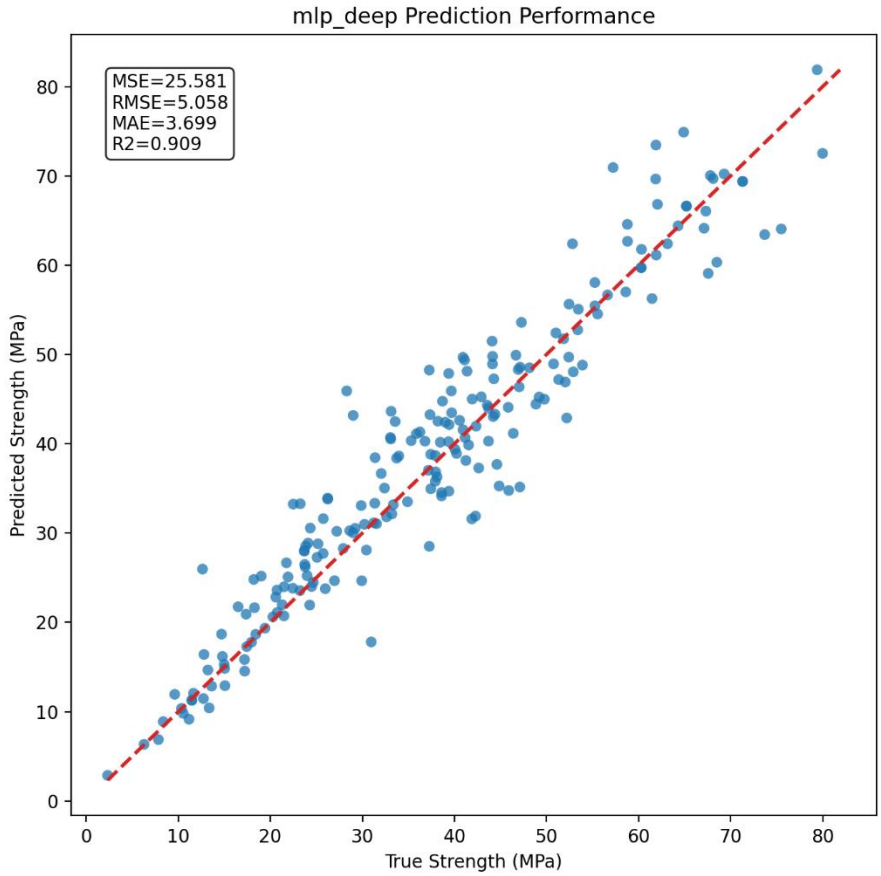
Mlp_small 的 loss 曲线与预测结果



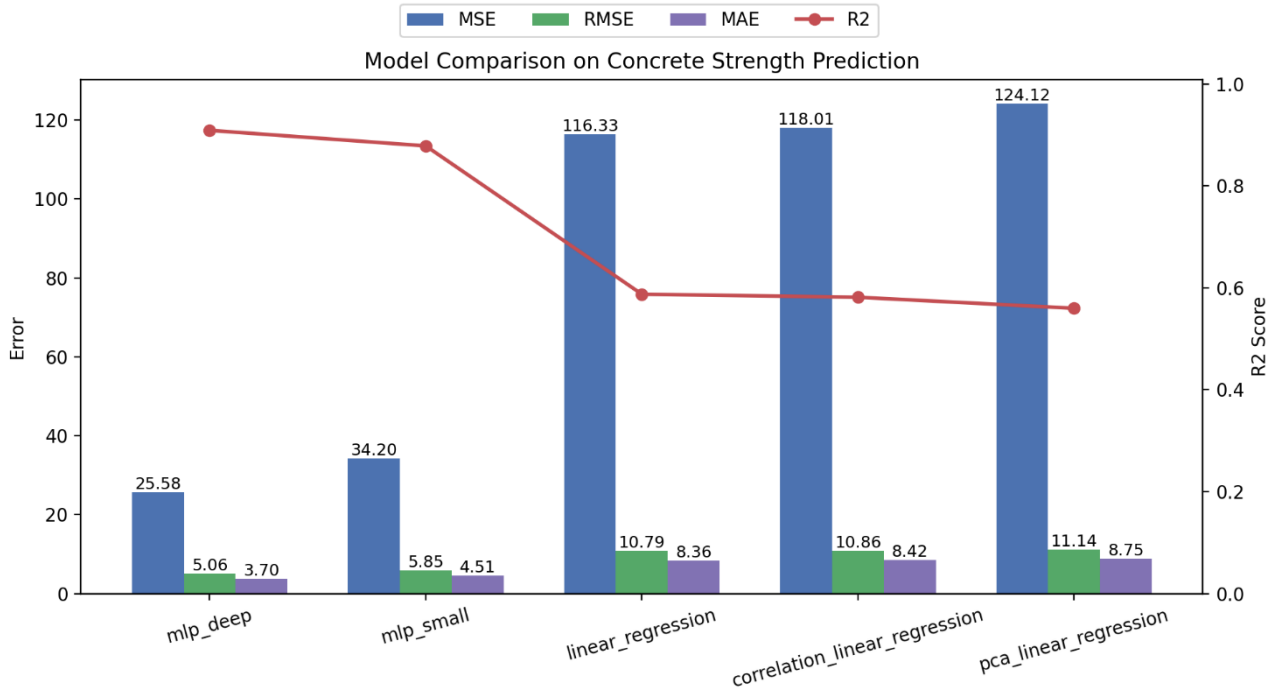


Mlp_deep 的 loss 曲线与预测结果





8、 实验结果



模型	MSE	RMSE	MAE	R2
MLP_DEEP	25.58	5.06	3.70	0.91
MLP_SMALL	34.20	5.85	4.51	0.88
LINEAR_REGRESSION	116.33	10.79	8.36	0.59
CORRELATION_LINEAR_REGRESSION	118.01	10.86	8.42	0.58
PCA_LINEAR_REGRESSION	124.12	11.14	8.75	0.56

实验结果分析：从实验结果来看，神经网络模型整体明显优于线性回归类模型，其中 MLP_DEEP 表现最好，MSE、RMSE、MAE 分别为 25.58、5.06、3.70，R^2 达到 0.91；MLP_SMALL 次之，R^2 为 0.88，也表现出较强的预测能力。相比之下，三种线性模型普通线性回归、基于相关性特征筛选的线性回归以及 PCA 降维后的线性回归结果接近，R^2 均仅在 0.56－0.59 左右，误差显著更大。这说明混凝土强度与输入特征之间可能存在较明显的非线性关系，单纯依赖线性模型难以充分刻画，而 MLP 能更好地学习这种复杂映射关系。此外，PCA 降维和简单相关性筛选并没有明显提升线性模型效果，甚至略有下降，表明在该任务中，保留原始特征并使用具有非线性建模能力的模型更为合适。